

JOÃO PAULO FONTOURA NOGUEIRA

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EDUCATION

MSc in Computer Science for Networks

Institut Polytechnique de Paris

Mention Très Bien (Highest Honors)

Sep. 2023 – Jul. 2025

Paris, France

- **Coursework:** Machine Learning; IoT & Data Analytics; Formal Verification; Network Science and Graph Learning; Optimization; Middleware for Distributed Application.

BSc in Electrical and Computer Engineering

Polytechnic Institute of Porto

ECTS Grading: A, Top 10%, Highest Honors

Sep. 2020 – Jul. 2023

Porto, Portugal

- **Coursework:** Applied AI (Ranked 1st); Systems Theory; Control Systems; Computer Vision; Signals and Systems; Advanced Computational Systems; Linear Algebra; Vibrations and Waves.

PUBLICATIONS

- **João Paulo Fontoura Nogueira**, Wentao Sun, Alonso Silva, Laith Zumot. “Certainty-Guided Reasoning in Large Language Models: A Dynamic Thinking Budget Approach.” *Under review at ICML 2026*. arXiv:2509.07820.
- Wentao Sun, **João Paulo Fontoura Nogueira**, Alonso Silva. “Structured Thinking Matters: Improving LLMs Generalization in Causal Inference Tasks.” *Under review at ICML 2026*. arXiv:2505.18034, 2025.

RESEARCH & PROFESSIONAL EXPERIENCE

Machine Learning Research Intern

Nokia Bell Labs

Apr 2025 – Sep 2025

Paris, France

- Proposed *Certainty-Guided Reasoning (CGR)*, an inference-time method for adaptive thinking budgets in reasoning LLMs, preserving accuracy while saving millions of tokens on AIME2024/2025.
- Engineered a scalable LLM training and evaluation pipeline using VLLM, managing multi-day runs on H100 GPU clusters; optimized workflows to resolve parallelization and data management challenges.
- Co-authored *Structured Thinking Matters*, achieving +47.5% relative F1 on Corr2Cause.
- Received the Outstanding Innovation Award from the Nokia Bell Labs Global Student Program; first France-based intern to receive this award.
- Accepted for oral presentation, LINC'S Annual Workshop, Palaiseau, France, June 2025.

Networks Research Intern

Institut Polytechnique de Paris

Nov 2023 – Jun 2024

Paris, France

- Implemented Cognitive Radio over LoRa networks enabling dynamic frequency switching on unlicensed spectrum.
- Developed MATLAB simulations and validated spectrum sensing and channel switching on SDR hardware.
- Cassiopée Award – Best Student Project 2024 (top among 60+ projects).

Robotics Research Intern

INESC TEC – Robotics and IoT Laboratory, University of Porto

Feb 2023 – Jun 2023

Porto, Portugal

- Developed a low-latency C driver on RP2040 for LD19 LiDAR, handling raw packet parsing and direct memory management
- Built a reliable LiDAR data pipeline to stream scans via UART to host devices, supporting autonomous navigation in agricultural field robots.
- Enhanced an Android (Java) application to generate a 360° 2D map from incoming LiDAR data.

SELECTED PROJECTS

LiDAR Mapping & SLAM Implementation	2025
· Implemented ICP scan registration and a simplified GraphSLAM back-end for 2D LiDAR mapping (U2IS/FR079 datasets), comparing against an incremental ICP-SLAM baseline.	
Autonomous Dragster Robot	2023
· Built an autonomous dragster to complete a 10 m sprint at max speed and stop precisely at the finish line. · Used infrared sensing and a PID-based controller for line/target detection and braking control.	
Real-time Facial Recognition	2023
· Fine-tuned YOLOv8m (Ultralytics/PyTorch) via transfer learning on a custom-labelled webcam dataset for closed-set identification of two individuals, rejecting others. · Ran live webcam inference with a high-confidence threshold; evaluated using confusion matrices and precision/recall/F1 curves.	
UWB Radar Sensing: Movement Tracking & Breath Monitoring	2023
· Implemented IR-UWB radar movement tracking in MATLAB. · Performed non-contact breath monitoring using envelope extraction (Hilbert transform) and variance-based feature selection, with Python analysis (NumPy/SciPy/Matplotlib) and time-/frequency-domain validation.	
Train Crowding Forecasting (SNCF-Transilien)	2024
· Built and compared ML models (Random Forest, Neural Networks, Polynomial Regression) to forecast real-time train occupancy using sensor-derived and lag features, handling missing and noisy data. · Achieved RMSE 0.0229 with Random Forest vs 0.158 baseline (84.6% improvement) through feature engineering from existing variables.	
Middleware & Digital Twin for Collision Avoidance	2024
· Created a digital twin simulation in C++ of cars on a road to predict and prevent vehicle collisions. · Used MQTT to connect clients (cars) and server via a pub-sub architecture.	
Embedded Tetris Table	2022
· Recreated Tetris game logic in C on an ATmega328P under tight memory constraints (2 KB RAM) · Controlled 300+ WS2812 RGB LEDs for real-time display output. · Developed a separate Android app to control the custom-built board via Bluetooth.	

AWARDS & DISTINCTIONS

- Outstanding Innovation Award, Nokia Bell Labs Global Student Program (2025)
- Mention Très Bien (Highest Honors) – MSc in Computer Science for Networks (2025)
- Cassiopée Award, Best Student Research Project – Institut Polytechnique de Paris (2024)
- Highest Honors, top 10% – BSc Electrical and Computer Engineering (2023)
- Ranked 1st in the selective Applied AI course at P. Porto, admission based on academic merit (2023)

SKILLS

Languages	Portuguese (native); English (native); French (B2); Spanish (B1); Italian (A1)
Programming	C/C++; Python; Assembly; Java; JavaScript; MATLAB
Libraries	PyTorch; PySMT, Keras; scikit-learn; OpenCV; vLLM; SimPy; NumPy; Pandas;
Tools	Linux; Git; Docker; Tinkercad; Google Cloud; SQLite; MQTT; PSpice; Ollama